Assignment 4

Q. Inheritance : it is an OOP Concept where a child class (subclass) acquires properties and methods from a parent class (super class) using the 'is-a' relationship, enabling code reuse and hierarchy.

Ex. i) Bank Account

++ Bank Account (Parent) ++
- accountHolderName : string
- accountNo : int
- CurrentBalance : double
+ deposit ()
+ withdraw ()
+ checkBalance ()

(Inherit to)

+ Saving Acc +
- minBalance : double
- interestrate : double
+ calculateInterest ()

+ Current Acc +
- Overdraft lim : double
- buisness Name : string
+ Overdraft ()

+ Loan Acc +
- LoanAmt : double
- emi Amt : double
+ PayEMI ()

ii) Vehicle (Base class)

++ Vehicle ++
- VehicleNo : string
- brand : string
- Speed : int
+ Start ()
+ stop ()
+ accelerate ()

+ Car +
- NumberofDoors : int
- acAvaible : boolean
+ Open_Trunk ()

+ Bike +
- Type : String
- hasSidebar : boolean
+ doWheelie ()

Page .....
Date .....

iii) Employee (Parent)

++ Employee ++
- employeeId : int
- name : String
- baseSalary : double
+ work()
+ attendMeeting()
+ getSalary()

* Manager *
- teamSize : int
- department : String
+ approveLeave()

* Developer *
- favColor : String
- isIndoor : boolean
+ writeCode()

iv) Animal (Base)

++ Animal ++
- name : String
- age : int
- weight : double
+ eat()
+ sleep()
+ breath()

* Dog *
- breed : String
- isVaccinated : boolean
+ bark()

* Cat *
- fur color : String
- isIndoor : boolean
+ purr()

v) Shape (Base)

++ Shape ++
- color : String
- borderThickness : int
- FillColor : String
+ draw()
+ resize()
+ getarea()

* Circle *
- radius : double
+ getCircumference()

* Rectangle *
- length : int
- width : int
+ getDiagonal()

CamScanner

vi) Person (Base)

++ Person ++
- name : string
- age : int
- add : string
+ eat ()
+ walk ()
+ talk ()

+ Student +
- rollNo : int
- course : string
+ study ()

+ Teacher +
- subject : string
- yearOfExp : int
+ teach ()

vii) Electronic Devices (Base)

++ Electronic Devices ++
- brand : string
- model : string
- price : double
+ PowerOn ()
+ PowerOff ()
+ reset ()

+ Smartphone +
- imei : string
- simType : string
+ makeCall ()

+ Laptop +
- ram : int
- storageSize : int
- runProgram ()

viii) Food Item (Base)

++ Food Item ++
- name : string
- calories : int
- expiryDate : date
+ Prepare ()
+ Serve ()
+ Store ()

+ Vegetarian +
- ingredient : List
+ checkVegCert ()

+ Non-Vegetarian +
- MeatType : String
+ checkFreshness ()

Page .....
Date .....

ix) Insuarance Policy (Parent)

++ Insurance Policy ++

-PolicyNo : int
-holderName : string
-PremiumAmount : double

+PayPremium()
+FileClaim()
+renewPolicy()

+ Heath Insurance +

-Coverage Amt : double
-hospitalNetwork : string
+ bookHospital()

+ Life Insurance +

- SumAssured : double
- nomineeName : string
+ addNominee()

x) Sports Person (Base)

++ Sports Person ++

-name : string
-age : int
-nationality : string

+ Practice()
+ Compete()
+ rest()

+ Cricketer +

-battingAvg : double
-bowlingStyle : string
+ bat()
+ bowl()

+ Footballer +

-Position : string
-goalScored : int
+ kick()
+ dribble()